

Fizika

Milliy sertifikat imtihoni topshiriqlari
29-sentabr 2026, 2-smena: shartlar va batafsil yechimlar

Online Fizika

Umumiy shart. Masalalarda berilmagan hollarda havo qarshiligi va yo'qotishlar hisobga olinmaydi, $g = 10 \text{ m/s}^2$.

Manba haqida. Topshiriqlar tartibi Azimov Doston (@smart_fizika) tuzgan 2-smena savollari to'plamiga mos keladi. Manbada javoblar kaliti yo'q edi, shuning uchun hamma javob mustaqil hisoblab chiqildi. Shartlar qayta bayon qilingan. 29–32-topshiriqlar manbada bo'sh qoldirilgan, shu sababli bu hujjatda yo'q. Rasmi masalalarda (11, 15, 21, 33–36) rasmdan o'qilgan ma'lumot shartda alohida yozilgan.

A qism. 1–28-topshiriqlar (to'g'ri javobni belgilang)

1-masala

1,8 km/min tezlikni m/s da ifodalang.

A) 1800 m/s B) 30 m/s C) 108 m/s D) 3 m/s

Yechim. $1,8 \text{ km/min} = \frac{1800 \text{ m}}{60 \text{ s}} = 30 \text{ m/s}$.

Javob: B) 30 m/s

2-masala

Jismga $F_1 = 60 \text{ N}$ va $F_2 = 80 \text{ N}$ kuchlar ta'sir qiladi. Natijaviy kuchning maksimal va minimal qiymatlari farqini toping.

A) 20 N B) 140 N C) 120 N D) 100 N

Yechim. Kuchlar bir yo'nalishda bo'lsa $R_{\max} = 60 + 80 = 140 \text{ N}$, qarama-qarshi bo'lsa $R_{\min} = 80 - 60 = 20 \text{ N}$.

$$R_{\max} - R_{\min} = 140 - 20 = 120 \text{ N}.$$

Javob: C) 120 N

3-masala

SOS signalining (radioto lqin) to'lqin uzunligi 600 nm. Uning chastotasini toping.

A) $5 \cdot 10^{14} \text{ Hz}$ B) $5 \cdot 10^5 \text{ Hz}$ C) $1,8 \cdot 10^{17} \text{ Hz}$ D) $2 \cdot 10^{-15} \text{ Hz}$

Yechim. $\nu = \frac{c}{\lambda} = \frac{3 \cdot 10^8}{600 \cdot 10^{-9}} = 5 \cdot 10^{14} \text{ Hz}$.

Javob: A) $5 \cdot 10^{14} \text{ Hz}$

4-masala

Gorizontal joylashgan kondensator plastinalari orasidagi kuchlanish 20 V bo'lganda zaryadlangan zarra muallaq turibdi. Kuchlanish 24 V ga yetkazilsa, zarra qanday tezlanish oladi?

A) 2 m/s^2 B) 12 m/s^2 C) $1,2 \text{ m/s}^2$ D) 4 m/s^2

Yechim. Muallaq turganda $q \frac{U_1}{d} = mg$. Kuchlanish $\frac{24}{20} = 1,2$ marta oshganda elektr kuch $1,2mg$ bo'ladi:

$$ma = 1,2mg - mg = 0,2mg \Rightarrow a = 0,2g = 2 \text{ m/s}^2 \text{ (yuqoriga).}$$

Javob: A) 2 m/s^2

5-masala

Suvda suzib yurgan muz bo'lagi ichida po'kak bor. Muz erisa, idishdagi suv sathi qanday o'zgaradi?

A) ko'tariladi B) pasayadi C) o'zgarmaydi D) po'kak massasiga bog'liq

Yechim. Muz va po'kak birga suzib yuribdi, demak siqib chiqarilgan suv hajmi $V_1 = \frac{m_{muz} + m_p}{\rho_s}$.

Muz erigach hosil bo'lgan suvning massasi m_{muz} bo'lib, uning hajmi $\frac{m_{muz}}{\rho_s}$; po'kak esa endi suvda suzadi va faqat o'z og'irligiga teng $\frac{m_p}{\rho_s}$ hajmdagi suvni siqib chiqaradi. Jami hajm o'zgarmaydi:

$$V_2 = \frac{m_{muz}}{\rho_s} + \frac{m_p}{\rho_s} = V_1.$$

Suv sathi o'zgarmaydi.

Javob: C) o'zgarmaydi

6-masala

Matematik mayatnikda uzunligi 90 cm bo'lgan ipga sharcha osilgan. Muvozanat vaziyatidan o'tayotgan paytda sharcha tezligi 0,15 m/s. Tebranish amplitudasini toping.

A) 4,5 cm B) 13,5 cm C) 9 cm D) 0,45 cm

Yechim. Muvozanat vaziyatida tezlik maksimal: $v_{max} = A\omega$, $\omega = \sqrt{g/l}$.

$$A = \frac{v_{max}}{\omega} = v_{max} \sqrt{\frac{l}{g}} = 0,15 \sqrt{\frac{0,9}{10}} = 0,15 \cdot 0,3 = 0,045 \text{ m} = 4,5 \text{ cm}.$$

Javob: A) 4,5 cm

7-masala

O'tkazgichning ko'ndalang kesimidan 2 min da 240 C zaryad o'tdi. Tok kuchini toping.

A) 2 A B) 120 A C) 4 A D) 0,5 A

Yechim. $I = \frac{q}{t} = \frac{240}{120} = 2 \text{ A}.$

Javob: A) 2 A

8-masala

Temperaturasi -20°C bo'lgan 500 g muzga 120 kJ issiqlik berildi. Muzning necha foizi eriydi? $c_{muz} = 2100 \text{ J/(kg} \cdot \text{K)}$, $\lambda = 330 \text{ kJ/kg}.$

A) 30 % B) 60 % C) 73 % D) 100 %

Yechim. Muzni 0°C gacha isitish: $Q_1 = cm\Delta T = 2100 \cdot 0,5 \cdot 20 = 21 \text{ kJ}.$

Erişhga qolgan issiqlik: $Q_2 = 120 - 21 = 99 \text{ kJ}$. Eriydigan massa:

$$m_e = \frac{Q_2}{\lambda} = \frac{99}{330} = 0,3 \text{ kg}, \quad \frac{m_e}{m} = \frac{0,3}{0,5} = 0,6 = 60 \%$$

Javob: B) 60 %

9-masala

Hajmi 3 l bo'lgan idishda 1,2 mol gaz bor. Gaz molekularining konsentratsiyasini toping. $N_A = 6 \times 10^{23} \text{ mol}^{-1}$.

A) $2,4 \cdot 10^{23} \text{ m}^{-3}$ B) $7,2 \cdot 10^{23} \text{ m}^{-3}$ C) $2,4 \cdot 10^{26} \text{ m}^{-3}$ D) $1,5 \cdot 10^{26} \text{ m}^{-3}$

Yechim. $n = \frac{N}{V} = \frac{\nu N_A}{V} = \frac{1,2 \cdot 6 \cdot 10^{23}}{3 \cdot 10^{-3}} = 2,4 \cdot 10^{26} \text{ m}^{-3}$.

Javob: C) $2,4 \cdot 10^{26} \text{ m}^{-3}$

10-masala

Ketma-ket ulangan ikkita elektrolitik vannadan birida Al^{3+} ionlari, ikkinchisida Cu^{2+} ionlari bo'lgan eritma bor. Ma'lum vaqt ichida ikkinchi vannada 48 mg mis ajraladi. Shu vaqtda birinchi vannada qancha alyuminiy ajraladi? $M(\text{Al}) = 27 \text{ g/mol}$, $M(\text{Cu}) = 64 \text{ g/mol}$.

A) 13,5 mg B) 20,25 mg C) 32 mg D) 170,7 mg

Yechim. Ketma-ket ulanganda ikkala vannadan bir xil zaryad o'tadi. Faradey qonuniga ko'ra $m = \frac{M}{zF} q$, ya'ni $m \propto \frac{M}{z}$:

$$\frac{m_{\text{Al}}}{m_{\text{Cu}}} = \frac{27/3}{64/2} = \frac{9}{32} \Rightarrow m_{\text{Al}} = 48 \cdot \frac{9}{32} = 13,5 \text{ mg}.$$

Javob: A) 13,5 mg

11-masala

Zanjirda uchta bir xil manba (har birining EYuK i $\mathcal{E}$, ichki qarshiligi r) berk halqa shaklida ketma-ket ulangan. Rasmga ko'ra uchala manba qutblari halqa bo'ylab bir xil yo'nalgan. 2-manba qutblari orasidagi kuchlanish qancha?

A) $\mathcal{E}$ B) 0 C) $\mathcal{E}/3$ D) $3\mathcal{E}$

Yechim. Manbalar bir xil yo'nalgan, shuning uchun EYuKlar qo'shiladi: $\mathcal{E}_{um} = 3\mathcal{E}$, ichki qarshiliklar ham ketma-ket: $R = 3r$ (tashqi qarshilik yo'q).

$$I = \frac{3\mathcal{E}}{3r} = \frac{\mathcal{E}}{r}.$$

2-manba qutblaridagi kuchlanish:

$$U_2 = \mathcal{E} - Ir = \mathcal{E} - \frac{\mathcal{E}}{r} r = 0.$$

Javob: B) 0

12-masala

Massasi 3,6 kg bo'lgan quruq o'tin to'liq yonganda qancha issiqlik ajraladi ($\text{kW} \cdot \text{h}$ da)? O'tinning solishtirma yonish issiqligi $q = 1 \times 10^7 \text{ J/kg}$.

A) $36 \text{ kW} \cdot \text{h}$ B) $10 \text{ kW} \cdot \text{h}$ C) $1 \text{ kW} \cdot \text{h}$ D) $3,6 \text{ kW} \cdot \text{h}$

Yechim. $Q = qm = 10^7 \cdot 3,6 = 3,6 \times 10^7 \text{ J}$. $1 \text{ kW} \cdot \text{h} = 3,6 \times 10^6 \text{ J}$ bo'lgani uchun

$$Q = \frac{3,6 \cdot 10^7}{3,6 \cdot 10^6} = 10 \text{ kW} \cdot \text{h}.$$

Javob: B) $10 \text{ kW} \cdot \text{h}$

13-masala

Ikki jismning harakat tenglamalari $x_1 = 2t$ va $x_2 = 30 - 3t - 4t^2$ (SI da). Harakat boshlangandan 3 s o'tgach jismlarning bir-biriga nisbatan tezligini toping.

A) 25 m/s B) 29 m/s C) 17 m/s D) 14 m/s

Yechim. Tezliklar koordinataning hosilasi: $v_1 = 2 \text{ m/s}$, $v_2 = -3 - 8t$. $t = 3 \text{ s}$ da $v_2 = -27 \text{ m/s}$. Jismlar qarama-qarshi yo'nalishda harakatlanadi:

$$v_{nis} = |v_1 - v_2| = |2 - (-27)| = 29 \text{ m/s}.$$

Javob: B) 29 m/s

14-masala

Obyektivning fokus masofasi 5 cm bo'lgan fotoapparat yordamida 200 m uzoqlikdagi binoning balandligi 5 mm bo'lgan tasviri hosil qilindi. Binoning balandligini toping.

A) 2 m B) 200 m C) 20 m D) 50 m

Yechim. Bino juda uzoqda ($d \gg F$), shuning uchun tasvir fokal tekislikda hosil bo'ladi: $f \approx F$. Kattalashtirish:

$$\Gamma = \frac{f}{d} = \frac{H'}{H} \Rightarrow H = H' \frac{d}{F} = 5 \text{ mm} \cdot \frac{200}{0,05} = 5 \times 10^{-3} \text{ m} \cdot 4000 = 20 \text{ m}.$$

Javob: C) 20 m

15-masala

Vodorod atomi uchta foton nurladi. Rasmga ko'ra: 1-foton $n = 4 \rightarrow n = 1$, 2-foton $n = 3 \rightarrow n = 2$, 3-foton $n = 3 \rightarrow n = 1$ o'tishlarda chiqqan. Sathlar energiyasi: $E_1 = -13,6$, $E_2 = -3,4$, $E_3 = -1,51$, $E_4 = -0,85 \text{ eV}$. Nurlangan foton impulslarini taqqoslang.

A) $p_2 > p_3 > p_1$ B) $p_3 > p_1 > p_2$ C) $p_1 > p_2 > p_3$ D) $p_1 > p_3 > p_2$

Yechim. Foton impulsi $p = \frac{E_f}{c}$, ya'ni foton energiyasiga proporsional. Energiyalari:

$$E_1^f = -0,85 - (-13,6) = 12,75 \text{ eV},$$

$$E_2^f = -1,51 - (-3,4) = 1,89 \text{ eV},$$

$$E_3^f = -1,51 - (-13,6) = 12,09 \text{ eV}.$$

$12,75 > 12,09 > 1,89$, demak $p_1 > p_3 > p_2$.

Javob: D) $p_1 > p_3 > p_2$

16-masala

$^{207}_{82}\text{Pb}$ qo'rg'oshin yadrosidagi neytronlar sonini aniqlang.

A) 207 B) 125 C) 82 D) 289

Yechim. $N = A - Z = 207 - 82 = 125$.

Javob: B) 125

17-masala

Massasi 1,75 kg bo'lgan jismga 3 N va 5 N kuchlar ta'sir etmoqda. Jism 4 m/s^2 tezlanish olsa, kuchlar orasidagi burchak qanday?

A) 30° B) 90° C) 60° D) 120°

Yechim. Teng ta'sir etuvchi kuch: $R = ma = 1,75 \cdot 4 = 7 \text{ N}$. Kosinuslar teoremasi:

$$R^2 = F_1^2 + F_2^2 + 2F_1F_2 \cos \alpha \Rightarrow 49 = 9 + 25 + 30 \cos \alpha \Rightarrow \cos \alpha = \frac{1}{2},$$

$\alpha = 60^\circ$.

Javob: C) 60°

18-masala

Liftga o'rnatilgan dinamometr 8 N ni ko'rsatmoqda. Unga ilingan yukning massasi 500 g bo'lsa, lift qanday tezlanish bilan harakatlanmoqda?

A) 16 m/s^2 B) 6 m/s^2 C) 4 m/s^2 D) 2 m/s^2

Yechim. Yukning og'irligi $mg = 0,5 \cdot 10 = 5 \text{ N}$, dinamometr ko'rsatishi (tortish kuchi) $T = 8 \text{ N}$. Yuqoriga yo'nalgan tezlanish bilan:

$$T - mg = ma \Rightarrow a = \frac{8 - 5}{0,5} = 6 \text{ m/s}^2 \text{ (yuqoriga).}$$

Javob: B) 6 m/s^2

19-masala

Massasi 2 kg, hajmi 4 dm^3 bo'lgan jism suv ostida 6 m chuqurlikda ushlab turilibdi. Jism qo'yib yuborilgandan keyin suv sirtidan 4 m balandlikda uning kinetik energiyasi qanday bo'ladi? Qarshilik kuchlarini hisobga olmang.

A) 120 J B) 80 J C) 200 J D) 40 J

Yechim. Jism zichligi $\rho = \frac{2}{0,004} = 500 \text{ kg/m}^3 < \rho_{suv}$, u yuqoriga qalqiydi. Suvdagi Arximed kuchi va og'irlik:

$$F_A = \rho_{suv} g V = 1000 \cdot 10 \cdot 0,004 = 40 \text{ N}, \quad mg = 20 \text{ N}.$$

Suv ichida (6 m): teng ta'sir etuvchi kuch $40 - 20 = 20 \text{ N}$ yuqoriga,

$$E_k = 20 \cdot 6 = 120 \text{ J}.$$

Suv sirtidan yuqorida (4 m): faqat og'irlik kuchi ishlaydi:

$$E'_k = 120 - mgh = 120 - 20 \cdot 4 = 40 \text{ J}.$$

(Jismning suvdan chiqish paytidagi qisqa oraliq hisobga olinmadi, jism o'lchamlari kichik deb qaraladi.)

Javob: D) 40 J

20-masala

Massasi 2 kg bo'lgan jism h balandlikdan 10 m/s boshlang'ich tezlik bilan tik yuqoriga otildi. U erishgan maksimal potensial energiya 160 J bo'lsa, h ni toping.

A) 3 m B) 8 m C) 5 m D) 13 m

Yechim. Maksimal balandlik (yer sathidan): $H = \frac{E_{p,max}}{mg} = \frac{160}{2 \cdot 10} = 8$ m. Otilgan nuqtadan yuqoriga ko'tarilish:

$$\Delta h = \frac{v_0^2}{2g} = \frac{100}{20} = 5 \text{ m},$$

demak $h = H - \Delta h = 8 - 5 = 3$ m.

Javob: A) 3 m

21-masala

Uzunligi 16 m, massasi 12 kg bo'lgan bir jinsli to'sin O tayanchga qo'yilgan (to'sin 8 ta teng bo'lakka bo'lingan). Rasmga ko'ra tayanch chap uchdan 2 bo'lak (4 m) masofada, to'sinning chap uchiga pastga F kuch qo'yilgan, o'ng uchiga ilingan dinamometr 25 N ko'rsatadi (yuqoriga tortadi). To'sin muvozanatda. F kuchning O nuqtaga nisbatan momentini aniqlang.

A) 300 N · m B) 180 N · m C) 780 N · m D) 45 N · m

Yechim. Har bir bo'lak 2 m. O dan: chap uchigacha 4 m, massa markazigacha (o'rtada) 4 m o'ngda, o'ng uchigacha 12 m. To'sin og'irligi $mg = 120$ N.

O ga nisbatan momentlar tengligi (soat miliga qarshi = soat mili bo'yicha):

$$M_F + M_{dinam} = M_{og'} \Rightarrow M_F + 25 \cdot 12 = 120 \cdot 4.$$

$$M_F = 480 - 300 = 180 \text{ N} \cdot \text{m}.$$

(Bunda $F = M_F/4 = 45$ N; 45 soni kuchning o'zi, momenti emas.)

Javob: B) 180 N · m

22-masala

Yopiq idishdagi gaz temperaturasi 4 marta pasayganda uning bosimi 60 kPa ga kamaydi. Gazning oxirgi bosimini toping.

A) 80 kPa B) 20 kPa C) 15 kPa D) 240 kPa

Yechim. Izoxorik jarayon: $p \propto T$, bosim ham 4 marta kamayadi: $p_2 = \frac{p_1}{4}$.

$$p_1 - \frac{p_1}{4} = \frac{3}{4}p_1 = 60 \Rightarrow p_1 = 80 \text{ kPa}, \quad p_2 = 20 \text{ kPa}.$$

Javob: B) 20 kPa

23-masala

Yassi kondensator qoplamalaridan biridagi zaryad miqdori 3 marta orttirilsa, kondensator sig'imi qanday o'zgaradi?

A) 3 marta ortadi B) 3 marta kamayadi C) 9 marta ortadi D) o'zgarmaydi

Yechim. Yassi kondensator sig'imi $C = \frac{\varepsilon \varepsilon_0 S}{d}$ faqat geometriyaga va muhitga bog'liq, zaryadga bog'liq emas (zaryad ortsa kuchlanish proporsional ortadi).

Javob: D) o'zgarmaydi

24-masala

Bir xil materialdan yasalgan ikkita prujinaning diametrlari $d_2 = 2d_1$, hajmlari $V_2 = 2V_1$. Ularning har ikkisi bir xil yuk osilsa, absolyut uzayishlari nisbati x_1/x_2 nimaga teng bo'ladi? (Prujinalar Guk qonuniga bo'ysunuvchi silindrsimon elastik jism kabi qaraladi.)

A) 1/8 B) 4 C) 8 D) 2

Yechim. Bir xil material, demak Yung moduli E bir xil. Jism ko'ndalang kesimi $S \propto d^2$, uzunligi $l = \frac{V}{S}$:

$$S_2 = 4S_1, \quad l_2 = \frac{V_2}{S_2} = \frac{2V_1}{4S_1} = \frac{l_1}{2}.$$

Bikrlik $k = \frac{ES}{l}$:

$$\frac{k_2}{k_1} = \frac{S_2/l_2}{S_1/l_1} = \frac{4}{1/2} = 8.$$

Bir xil yuk uchun $x = \frac{F}{k}$:

$$\frac{x_1}{x_2} = \frac{k_2}{k_1} = 8.$$

Javob: C) 8

25-masala

4 m balandlikkacha suv quyilgan idish tubidagi yuzasi 30 cm^2 bo'lgan teshik tiqin bilan berkitilgan. Tiqinga ta'sir qiluvchi gidrostatik bosim kuchini aniqlang.

A) 120 N B) 12 N C) 1200 N D) 80 N

Yechim. $p = \rho gh = 1000 \cdot 10 \cdot 4 = 40\,000 \text{ Pa}$, $S = 30 \text{ cm}^2 = 3 \times 10^{-3} \text{ m}^2$:

$$F = pS = 40000 \cdot 3 \cdot 10^{-3} = 120 \text{ N}.$$

Javob: A) 120 N

26-masala

Induktivligi 12 mH bo'lgan g'altakdagi tok kuchi 2 A ga ortganda undagi magnit maydon energiyasi 48 mJ ga ortdi. G'altakdagi oxirgi magnit oqimini toping.

A) 24 mWb B) 36 mWb C) 12 mWb D) 48 mWb

Yechim. $I_2 = I_1 + 2$. Energiya o'zgarishi:

$$\Delta W = \frac{L}{2} (I_2^2 - I_1^2) = \frac{L}{2} (I_2 - I_1)(I_2 + I_1) \Rightarrow 0,048 = 0,006 \cdot 2 \cdot (I_1 + I_2),$$

$$I_1 + I_2 = 4, \quad I_2 - I_1 = 2 \Rightarrow I_2 = 3 \text{ A}.$$

Oxirgi magnit oqim:

$$\Phi = LI_2 = 0,012 \cdot 3 = 0,036 \text{ Wb} = 36 \text{ mWb}.$$

Javob: B) 36 mWb

27-masala

Monoxromatik yorug'lik davri $d = 3 \mu\text{m}$ bo'lgan difraksion panjaraga tik tushmoqda. Yorug'likning to'lqin uzunligi 600 nm. Kuzatilishi mumkin bo'lgan eng katta maksimum tartibi

qanday?

A) 5 B) 4 C) 3 D) 10

Yechim. Maksimum sharti $d \sin \varphi = k\lambda$, $\sin \varphi < 1$ bo'lishi kerak:

$$k < \frac{d}{\lambda} = \frac{3000}{600} = 5.$$

$k = 5$ da $\sin \varphi = 1$ ($\varphi = 90^\circ$), bunday maksimum ekranda kuzatilmaydi. Eng katta kuzatiladigan tartib $k = 4$.

Javob: B) 4**28-masala**

Elektronning impulsi to'liq uzunligi 490 nm bo'lgan foton impulsiga teng. Elektronning tezligini toping ($h = 6,63 \times 10^{-34} \text{ J} \cdot \text{s}$, $m_e = 9,1 \times 10^{-31} \text{ kg}$).

A) 3,6 km/s B) 1,5 km/s C) 0,74 km/s D) 1,5 m/s

Yechim. $m_e v = \frac{h}{\lambda}$:

$$v = \frac{h}{m_e \lambda} = \frac{6,63 \cdot 10^{-34}}{9,1 \cdot 10^{-31} \cdot 4,9 \cdot 10^{-7}} \approx 1,49 \cdot 10^3 \text{ m/s} \approx 1,5 \text{ km/s}.$$

Javob: B) 1,5 km/s**B qism. 33–35-topshiriqlar (moslashtirish)****33–35-masala**

Zanjir bitta berk konturdan iborat: ketma-ket E_2 (r_2), E_1 (r_1) manbalar, R_3 rezistor, ideal ampermetr va o'zaro parallel ulangan R_1, R_2 (unga ideal voltmeter parallel). Rasmga ko'ra ikkala manbaning musbat qutbi yuqorida. $R_1 = 2 \Omega$, $R_2 = 6 \Omega$, $R_3 = 8,5 \Omega$, $r_1 = 3,5 \Omega$, $r_2 = 1,5 \Omega$, $\mathcal{E}_1 = 80 \text{ V}$, $\mathcal{E}_2 = 20 \text{ V}$.

33. Ampermetr ko'rsatishi (A). 34. Voltmetr ko'rsatishi (V). 35. 0,1 s da ikkinchi manba energiyasining o'zgarishi (J).

A) 4 B) 1,5 C) 8 D) 6 E) 12 F) 2

Yechim. Ikkala manbaning musbat qutbi yuqorida, halqa bo'ylab yurganda ular bir-biriga *qarama-qarshi* ulangan, shuning uchun EYuKlar ayriladi (kuchliroq $\mathcal{E}_1$ tok yo'nalishini belgilaydi):

$$\mathcal{E} = \mathcal{E}_1 - \mathcal{E}_2 = 80 - 20 = 60 \text{ V}.$$

Parallel qism qarshiligi: $R_{12} = \frac{2 \cdot 6}{2 + 6} = 1,5 \Omega$. Umumiy qarshilik:

$$R = R_{12} + R_3 + r_1 + r_2 = 1,5 + 8,5 + 3,5 + 1,5 = 15 \Omega.$$

$$33. I = \frac{\mathcal{E}}{R} = \frac{60}{15} = 4 \text{ A} \Rightarrow \text{A}.$$

$$34. \text{ Voltmetr } R_{12} \text{ dagi kuchlanishni ko'rsatadi: } U = IR_{12} = 4 \cdot 1,5 = 6 \text{ V} \Rightarrow \text{D}.$$

35. Tok E_2 manbaning musbat qutbidan ichkariga kiradi, ya'ni ikkinchi manba zaryadlanmoqda va uning energiyasi ortadi:

$$\Delta W_2 = \mathcal{E}_2 It = 20 \cdot 4 \cdot 0,1 = 8 \text{ J} \Rightarrow \text{C}.$$

Javob: 33 – A; 34 – D; 35 – C

C qism. 36–45-topshiriqlar (javobni yozing)

36-masala

Vaznsiz blok orqali o'tkazilgan ipning bir uchiga 10 kg, ikkinchi uchiga 5 kg yuk osilgan. Blok 200 N kuch bilan vertikal yuqoriga ko'tarilmoqda.

- a) Blokning ilgarilanma tezlanishini toping.
b) 10 kg li yukning Yerga nisbatan tezlanishini toping.

Yechim. Blok vaznsiz, shuning uchun unga ta'sir etuvchi kuchlar muvozanatda: $2T = F \Rightarrow T = 100 \text{ N}$.

10 kg li yuk: $T - m_1g = 100 - 100 = 0 \Rightarrow a_1 = 0$.

5 kg li yuk: $T - m_2g = 100 - 50 = 50 \text{ N} \Rightarrow a_2 = \frac{50}{5} = 10 \text{ m/s}^2$ (yuqoriga).

Ip uzunligi o'zgarmaydi: $y_1 + y_2 - 2y_b = \text{const}$, demak blok tezlanishi

$$a_b = \frac{a_1 + a_2}{2} = \frac{0 + 10}{2} = 5 \text{ m/s}^2 \text{ (yuqoriga).}$$

Javob: a) $a_b = 5 \text{ m/s}^2$; b) $a_1 = 0$ (10 kg li yuk Yerga nisbatan tinch turadi)

37-masala

Qoplamalari orasidagi masofa 4 mm bo'lgan kondensatorga dielektrik singdiruvchanligi 2 ga teng modda kiritildi va u 80 V kuchlanishga ulandi ($\epsilon_0 = 8,85 \times 10^{-12} \text{ F/m}$).

- a) Elektr maydon kuchlanganligini toping (kV/m).
b) Elektr maydon energiyasining zichligini toping.

Yechim. a) Kuchlanish manbaga ulangan, shuning uchun

$$E = \frac{U}{d} = \frac{80}{4 \cdot 10^{-3}} = 2 \times 10^4 \text{ V/m} = 20 \text{ kV/m}.$$

b)

$$w = \frac{\epsilon \epsilon_0 E^2}{2} = \frac{2 \cdot 8,85 \cdot 10^{-12} \cdot (2 \cdot 10^4)^2}{2} = 8,85 \cdot 10^{-12} \cdot 4 \cdot 10^8 \approx 3,5 \cdot 10^{-3} \text{ J/m}^3.$$

Javob: a) $E = 20 \text{ kV/m}$; b) $w \approx 3,5 \cdot 10^{-3} \text{ J/m}^3$

38-masala

$0,8c$ tezlik bilan harakatlanayotgan jismning impulsi $4 \times 10^{-20} \text{ kg} \cdot \text{m/s}$ ga teng ($c = 3 \times 10^8 \text{ m/s}$).

- a) Jismning to'la energiyasi qancha (pJ)?
b) Jismning kinetik energiyasi qancha (pJ)?

Yechim. Relyativistik: $p = \gamma mv$, $E = \gamma mc^2$, bundan $E = \frac{pc^2}{v}$.

a)

$$E = \frac{pc^2}{0,8c} = \frac{pc}{0,8} = \frac{4 \cdot 10^{-20} \cdot 3 \cdot 10^8}{0,8} = 1,5 \times 10^{-11} \text{ J} = 15 \text{ pJ}.$$

b) $\gamma = \frac{1}{\sqrt{1 - 0,64}} = \frac{5}{3}$. Tinchlikdagi energiya $E_0 = \frac{E}{\gamma} = 15 \cdot 0,6 = 9 \text{ pJ}$.

$$E_k = E - E_0 = 15 - 9 = 6 \text{ pJ}.$$

Javob: a) $E = 15 \text{ pJ}$; b) $E_k = 6 \text{ pJ}$

39-masala

Karno sikli bo'yicha ishlovchi mashinaning isitkich temperaturasi sovitkichnikidan 4 marta katta. Mashina bir siklda 30 kJ ish bajaradi.

- a) Mashinaning foydali ish koeffitsiyenti necha foiz?
b) 2 siklda sovitkichga berilgan issiqlik miqdori qancha?

Yechim. a)

$$\eta = 1 - \frac{T_s}{T_i} = 1 - \frac{1}{4} = 0,75 = 75 \%$$

- b) Bir sikl uchun isitkichdan olingan issiqlik $Q_i = \frac{A}{\eta} = \frac{30}{0,75} = 40 \text{ kJ}$, sovitkichga berilgani $Q_s = Q_i - A = 40 - 30 = 10 \text{ kJ}$. Ikki siklda:

$$2Q_s = 20 \text{ kJ}.$$

Javob: a) $\eta = 75 \%$; b) 20 kJ

40-masala

Buyumdan ekrangacha masofa 2,4 m. Ekranda buyumning 5 marta kattalashgan haqiqiy tasviri hosil bo'ldi.

- a) Buyumdan linzagacha masofa qancha?
b) Linzaning optik kuchi qancha?

Yechim. Haqiqiy tasvir uchun $\Gamma = \frac{f}{d} = 5 \Rightarrow f = 5d$, va $d + f = 2,4 \text{ m}$.

- a) $6d = 2,4 \Rightarrow d = 0,4 \text{ m} = 40 \text{ cm}$, $f = 2 \text{ m}$.

- b) Linza formulasi:

$$D = \frac{1}{F} = \frac{1}{d} + \frac{1}{f} = \frac{1}{0,4} + \frac{1}{2} = 2,5 + 0,5 = 3 \text{ dptr}.$$

Javob: a) $d = 0,4 \text{ m}$ (40 cm); b) $D = 3 \text{ dptr}$ ($F \approx 33,3 \text{ cm}$)

41-masala

Neft stansiyasida nasos har minutda $2,4 \text{ m}^3$ hajmdagi neftni ko'ndalang kesim yuzi $0,8 \text{ dm}^2$ bo'lgan quvur orqali 48 m balandlikka ko'taradi. Neftning zichligi 800 kg/m^3 .

- a) Neftning quvurdagi tezligi qancha?
b) Nasosning minimal quvvati qancha?

Yechim. a) Hajmiy sarf: $Q = \frac{2,4}{60} = 0,04 \text{ m}^3/\text{s}$, $S = 0,8 \text{ dm}^2 = 8 \times 10^{-3} \text{ m}^2$:

$$v = \frac{Q}{S} = \frac{0,04}{8 \cdot 10^{-3}} = 5 \text{ m/s}.$$

- b) Har sekunda massa $\mu = \rho Q = 800 \cdot 0,04 = 32 \text{ kg/s}$. Minimal quvvat (yo'qotishlarsiz) neftni h ga ko'tarishga va unga v tezlik berishga sarflanadi:

$$P = \mu \left(gh + \frac{v^2}{2} \right) = 32 \cdot \left(10 \cdot 48 + \frac{25}{2} \right) = 32 \cdot 492,5 = 15760 \text{ W} \approx 15,8 \text{ kW}.$$

(Faqat ko'tarishga sarf $\mu gh = 15,36 \text{ kW}$, kinetik energiya qismi $0,4 \text{ kW}$.)

Javob: a) $v = 5 \text{ m/s}$; b) $P_{\min} \approx 15,8 \text{ kW}$

42-masala

Solenoid ichidagi magnit maydon energiyasining hajmiy zichligi $2,4 \times 10^{-4} \text{ J/m}^3$, ko'ndalang kesim yuzi 4 cm^2 , uzunligi 50 cm , induktivligi $3 \times 10^{-7} \text{ H}$. $\pi \approx 3$ deb oling ($\mu_0 = 4\pi \cdot 10^{-7} \approx 1,2 \cdot 10^{-6} \text{ H/m}$).

- a) Solenoiddagi tok kuchini toping.
b) Magnit maydon kuchlanganligini toping.

Yechim. a) Solenoid hajmi $V = Sl = 4 \cdot 10^{-4} \cdot 0,5 = 2 \times 10^{-4} \text{ m}^3$. Magnit maydon energiyasi:

$$W = wV = 2,4 \cdot 10^{-4} \cdot 2 \cdot 10^{-4} = 4,8 \times 10^{-8} \text{ J}.$$

$$W = \frac{LI^2}{2} \text{ dan}$$

$$I = \sqrt{\frac{2W}{L}} = \sqrt{\frac{9,6 \cdot 10^{-8}}{3 \cdot 10^{-7}}} = \sqrt{0,32} \approx 0,57 \text{ A}.$$

b) Energiya zichligi $w = \frac{\mu_0 H^2}{2}$:

$$H = \sqrt{\frac{2w}{\mu_0}} = \sqrt{\frac{4,8 \cdot 10^{-4}}{1,2 \cdot 10^{-6}}} = \sqrt{400} = 20 \text{ A/m}.$$

(Mos ravishda $B = \mu_0 H = 24 \text{ mT}$.)

Javob: a) $I = \sqrt{0,32} \approx 0,57 \text{ A}$; b) $H = 20 \text{ A/m}$

43-masala

Yung moduli 180 GPa bo'lgan sterjenning boshlang'ich uzunligi 4 m , ko'ndalang kesim yuzi 12 mm^2 . Unga tashqi kuch qo'yilganda sterjen 80 cm ga cho'zildi.

- a) Nisbiy uzayishni toping (%).
b) Tashqi kuchni toping (kN).

Yechim. a)

$$\varepsilon = \frac{\Delta l}{l_0} = \frac{0,8}{4} = 0,2 = 20 \text{ \%}.$$

b) Guk qonuni: $\sigma = E\varepsilon$, $F = \sigma S$:

$$F = E\varepsilon S = 180 \cdot 10^9 \cdot 0,2 \cdot 12 \cdot 10^{-6} = 4,32 \times 10^5 \text{ N} = 432 \text{ kN}.$$

Javob: a) $\varepsilon = 20 \text{ \%}$; b) $F = 432 \text{ kN}$

44-masala

Jism 5 s da 3 m/s^2 tezlanish bilan tekis tezlanuvchan harakatlanib, 70 m masofani bosib o'tdi.

- a) Jismning boshlang'ich tezligini toping.
b) Shu 5 s dan oldingi 2 s da jism qancha yo'l bosib o'tgan? (Tezlanish o'zgarmagan.)

Yechim. a) $s = v_0 t + \frac{at^2}{2}$:

$$70 = 5v_0 + \frac{3 \cdot 25}{2} = 5v_0 + 37,5 \Rightarrow v_0 = \frac{32,5}{5} = 6,5 \text{ m/s}.$$

b) Oldingi 2 s boshidagi tezlik: $v = v_0 - at = 6,5 - 3 \cdot 2 = 0,5 \text{ m/s}$. Shu 2 s da yo'l:

$$s' = 0,5 \cdot 2 + \frac{3 \cdot 2^2}{2} = 1 + 6 = 7 \text{ m}.$$

Javob: a) $v_0 = 6,5 \text{ m/s}$; b) $s' = 7 \text{ m}$

45-masala

Fotoelementga chastotasi 4×10^{15} Hz bo'lgan fotonlar oqimi tushmoqda. Chastota 3 marta orttirilsa, to'xtatuvchi kuchlanish 4 marta ortadi. $h = 6,63 \times 10^{-34}$ J·s, $c = 3 \times 10^8$ m/s, $e = 1,6 \times 10^{-19}$ C.

a) Ushbu modda uchun fotoeffektning qizil chegarasini toping (nm).

b) Keyingi holatdagi fotoelektronlarning kinetik energiyasini toping (eV).

Yechim. $h\nu = A + eU_z$ (A – chiqish ishi). Birinchi holat: $h\nu_1 - A = eU_1$. Ikkinchi holat ($\nu_2 = 3\nu_1$, $U_2 = 4U_1$): $3h\nu_1 - A = 4eU_1$. Tenglamalarni bo'lamiz:

$$3h\nu_1 - A = 4(h\nu_1 - A) \Rightarrow 3A = h\nu_1 \Rightarrow A = \frac{h\nu_1}{3}.$$

a) Qizil chegara chastotasi $\nu_{min} = \frac{A}{h} = \frac{\nu_1}{3} = 1,33 \times 10^{15}$ Hz:

$$\lambda_{max} = \frac{c}{\nu_{min}} = \frac{3 \cdot 10^8}{1,33 \cdot 10^{15}} = 2,25 \times 10^{-7} \text{ m} = 225 \text{ nm}.$$

b) $h\nu_1 = 6,63 \cdot 10^{-34} \cdot 4 \cdot 10^{15} = 2,65 \times 10^{-18} \text{ J} \approx 16,6 \text{ eV}$. Ikkinchi holatda

$$E_k = 3h\nu_1 - A = \frac{8}{3} h\nu_1 = \frac{8}{3} \cdot 16,6 \approx 44 \text{ eV}.$$

Javob: a) $\lambda_{max} = 225 \text{ nm}$; b) $E_k \approx 44 \text{ eV}$

Javoblar kaliti

1–28-topshiriqlar

1	2	3	4	5	6	7	8	9	10
B	C	A	A	C	A	A	B	C	A
11	12	13	14	15	16	17	18	19	20
B	B	B	C	D	B	C	B	D	A
21	22	23	24	25	26	27	28		
B	B	D	C	A	B	B	B		

33–35-topshiriqlar

33	34	35
A	D	C

36–45-topshiriqlar

T/r	a)	b)
36	5 m/s^2	0
37	20 kV/m	$\approx 3,5 \cdot 10^{-3} \text{ J/m}^3$
38	15 pJ	6 pJ
39	75 %	20 kJ
40	0,4 m	3 dptr
41	5 m/s	$\approx 15,8 \text{ kW}$
42	$\approx 0,57 \text{ A}$	20 A/m
43	20 %	432 kN
44	$6,5 \text{ m/s}$	7 m
45	225 nm	$\approx 44 \text{ eV}$